

Blacklegged Ticks in Toronto Parks Are Concentrated in the East*

Three-quarters of surveys from 2013 to 2023 found no ticks, while five Rouge Valley sites produced two-thirds of the ticks collected and most of the infected ones

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We analyze Toronto Public Health’s surveillance of blacklegged ticks, which carry the bacterium that causes Lyme disease, using 233 surveys of 84 sites in city parks from 2013 to 2023. Three-quarters of surveys found no ticks, while five Rouge Valley sites, in the city’s northeast, produced 65% of the 606 ticks collected and 83% of those that tested positive. The rise in infected ticks after 2015 reflects how much of each year’s catch came from the Rouge Valley, not a change across Toronto. Knowing that Lyme disease risk is concentrated in a handful of parks can help focus prevention where it is needed most.

1 Introduction

As summers get warmer and the threat of climate change looms, Canadians have yet another thing to worry about: Lyme disease. But how big an issue is it, and is climate change actually linked to it? Lyme disease is becoming more common in Canada, with reported cases rising from 338 in 2012 to 2,525 in 2022 (Wafy et al. 2026) and preliminary counts passing 5,200 in 2024 (Dev 2026). The disease is spread by the blacklegged tick, also known as the deer tick, which carries the bacterium that causes it. Over recent decades, the blacklegged tick’s range and numbers have continued to grow (Eisen et al. 2016), and Ontario is at the centre of this change: in the first half of 2026, the province accounted for nearly two-thirds of blacklegged tick reports submitted by the public in Canada (Otis 2026). All of Toronto is now considered an estimated risk area for Lyme disease (Toronto Public Health 2026c). However, national surveillance reports only by province, and public reports depend on who notices a tick, if anyone does, so neither can say where within a city the risk lies.

*Code and data are available at: https://github.com/UpCatch/Toronto_Tick_Surveillance_Data.

In this paper, we analyze all 233 surveys in Toronto Public Health’s Blacklegged Tick Surveillance dataset, covering 84 sites between 2013 and 2023. Three-quarters of surveys found no ticks, and 52 of the 84 sites never recorded one. Most ticks were concentrated in a few places in the east of the city. Five sites in the Rouge Valley produced 65% of the 606 ticks collected, and a smaller cluster appeared on the Toronto Islands. Infected ticks were even more concentrated. 45% of Rouge Valley ticks tested positive, compared with 17% elsewhere, and these five sites held 83% of all positive ticks. The citywide share of positive ticks rose after 2015, but this reflects how much of each year’s catch came from the Rouge Valley rather than a trend across the city, and the data cannot show whether climate change played a role. This matters because surveillance is meant to show where and when to focus prevention (Wafy et al. 2026). It also shapes how the data should be read: the program samples in the fall and so largely misses nymphs, the tick stage most likely to infect people (Eisen et al. 2016).

The remainder of this paper is structured as follows. Section 2 describes the data source, how ticks are collected and tested, each variable in the dataset, and the patterns in tick counts, locations and infection. Section A describes how the raw data were cleaned and tested.

2 Data

2.1 Overview

We use the statistical programming language R (R Core Team 2026). The dataset used in this paper comes from Toronto Public Health’s Blacklegged Tick Surveillance dataset (Toronto Public Health 2026a), which was obtained from the City of Toronto’s Open Data Portal (Gelfand 2025). Additional packages in R that helped to support this paper include `tidyverse` (Wickham et al. 2019), `ggplot2` (Wickham 2016) and `tinytable` (Arel-Bundock 2026) for graphs and tables respectively, `here` (Müller 2025) for handling file paths, and the map for the City of Toronto’s municipal boundary (City of Toronto 2026), which `sf` (Pebesma 2018) helped to draw. With inspiration from Rohan Alexander’s book (Alexander 2023), we focus on what the data can tell us about the world we inhabit.

In Ontario, blacklegged ticks (*Ixodes scapularis*) are the only ticks that transmit *Borrelia burgdorferi*, the bacterium that causes Lyme disease (Toronto Public Health 2026c). Over the last two decades, Lyme disease in Canada and in the United States has increased both in incidence and geographic extent (Eisen et al. 2016). Eisen et al. explain that a tick’s development, survival, and its ability to find a host are directly affected by temperature and humidity. Compared to a generalized life cycle of approximately 2 years for ticks inhabiting the northeastern United States, the life cycle of some ticks in comparatively colder southeastern Canada may take 3 to 4 years, which lowers their reproductive rate (Eisen et al. 2016). Justin Wood, founder of the tick-testing company Geneticks, told CP24 that climate change is driving the rise in tick reports. Ontario accounted for 63.7% of blacklegged tick reports across Canada in the first half of 2026, and the province’s reports rose 72.8% from the same period in 2025

(Otis 2026). Toronto is now considered to be an Estimated Risk Area for Lyme disease, but Toronto Public Health considers the risk of infection low, although increasing (Toronto Public Health 2026c). National reports note that regular surveillance activity helps identify any shifting trends in Lyme disease occurrence and geographic distribution (Wafy et al. 2026).

Every row in the dataset represents a single site surveyed in a year. In total, there are 233 rows detailing 84 site names covered from 2013 to 2019 and 2023. No surveillance was conducted from 2020 to 2022. For each site and year in the dataset, Toronto Public Health records:

- **Site name** and **location** (as a single latitude and longitude representing the entire site)
- **Year** the surveillance took place
- **Adult and nymph ticks collected** (the last two stages in a blacklegged tick’s life cycle)
- **Larvae collected** (newly hatched ticks, the first stage in the life cycle)
- **Ticks that tested positive** (assumed to be for *Borrelia burgdorferi*; see below)

There is another column in the dataset named “Total BLTs”, but given that it is identical to the adult and nymph counts in all rows, we treat the two columns as one variable. Furthermore, the **Rouge Valley**, located on the Northeastern side of Toronto, contains 5 sites including the name “Rouge”. To learn more about the data cleaning process for this dataset, such as how the latitudes and longitudes were extracted, and how sites with similar names were handled, please see Section A.

2.2 Measurement

Toronto Public Health’s method of surveillance is a process termed “tick dragging”, involving a surveyor dragging a white flannel cloth behind them across the ground and through vegetation, waiting for ticks to latch on (Russell and Jain-Sheehan 2015). Sites are chosen according to how suitable they are for blacklegged ticks to inhabit or if there was a previously confirmed finding, and drags are conducted during the fall when adult ticks are at their most active (Toronto Public Health 2026b). Collected ticks are removed, placed into a collection vial, and shipped to the National Microbiology Laboratory to identify species and run diagnostics (Russell and Jain-Sheehan 2015). The results for every site then make up one row for that particular year.

While the dataset does not explicitly name the pathogen identified in the positive count column, Toronto Public Health states that testing is done for *B. burgdorferi*, and as of 2023, for the *Babesia* species, *Anaplasma phagocytophilum* and the Powassan virus. Given that the surveillance program started with the goal of tracking blacklegged ticks carrying Lyme disease as its core focus, we assume that the positive count values represent ticks with the *B. burgdorferi* bacteria present.

Each step of the method shapes what the data mean. Firstly, the count of blacklegged ticks at a given site is a sample rather than a census. Tick dragging fails to account for ticks attached to animals, and a drag may not “catch” every single tick along the path. The number of ticks found can also depend on the weather or the care of the surveyor. Toronto Public Health notes

that there may be ticks present at a site even if none were reportedly collected (Toronto Public Health 2026b). Secondly, the sites are chosen based on past tick findings or their suitability for ticks to inhabit. This choice, while understandable, can have the effect of overrepresenting sites with ticks. Lastly, dragging in the fall misses the peak of nymph activity, which runs from late May to early July (Eisen et al. 2016). Dragging is timed for adults partly because nymphs are so small (Russell and Jain-Sheehan 2015), but nymphs matter most for people: they are more likely than adults to stay attached long enough to transmit infection (Eisen et al. 2016).

2.3 Ethical considerations

The primary ethical consideration is that naming parks where ticks were found might prevent people from visiting them. This is possibly made worse by the fact that most sites with large tick presence are in Scarborough, in Toronto’s east end. Furthermore, a site not recording any ticks does not confirm that no ticks are present (Toronto Public Health 2026b). The dataset may be interpreted by the public as a ranking of which sites are safe to visit and which are not, when it should be perceived as which sites require the greatest amount of precaution.

2.4 Similar datasets

The Public Health Agency of Canada publishes a national summary once annually that combines tick counts from dragging with images of ticks submitted by the public, primarily through the online, image-based platform eTick.ca (Wafy et al. 2026). Otis notes that because eTick relies on users to submit data, confirmed numbers likely underestimate actual population numbers in Canada (Otis 2026). Accurate numbers from public submissions also rely on individuals noticing a tick and choosing to report it, which is not always the case. Toronto Public Health’s dataset was the only source found that records systematic tick collection at named locations in the city.

2.5 Summary of the variables

Table 1 takes the three count variables and provides a summary across all site visits for every year surveillance occurred. The mean represents the average tick tally per survey, the median is the middle value when surveys are ordered by count in ascending order, and the standard deviation shows the degree to which the counts are spread around the mean. The median is 0 for all variables, and the standard deviation is considerably higher than the mean in all cases, which means most surveys found nothing while a few found many ticks.

Table 2 showcases the same counts broken down by year. From the table, we can see that the number of sites surveyed went from 1 in 2013 to 36 in 2016. This suggests that a year with a greater tick count can be a result of the increase in sites searched. Thus, yearly totals are not a good indicator of the tick count in Toronto.

Table 1: Summary statistics for tick counts across all 233 surveys (one per site per year).

Variable	Mean	Median	Standard deviation	Maximum	Total	Surveys above zero
Adults and nymphs	2.60	0	11.24	122	606	57
Larvae	0.09	0	0.85	9	22	4
Tested positive	0.91	0	5.25	63	211	19

Table 2: Sites surveyed, ticks collected and ticks that tested positive by year, 2013–2023.

Year	Parks surveyed	Parks with ticks	Adults and nymphs	Larvae	Tested positive	Positive (%)
2013	1	1	1	0	0	0.0
2014	4	1	3	0	0	0.0
2015	27	4	17	0	1	5.9
2016	36	6	95	0	30	31.6
2017	38	8	150	11	65	43.3
2018	46	12	96	11	36	37.5
2019	39	17	123	0	36	29.3
2023	42	8	121	0	43	35.5

2.6 Most surveys found no ticks

Looking at all 233 surveys, 176 (76%) of them found no blacklegged ticks (Figure 1). While the average survey found 2.6 ticks, this is inflated by some sites having much larger counts (e.g., Rouge Park in 2017 with 122 ticks found). Over all surveys, 606 adults and nymphs were collected.

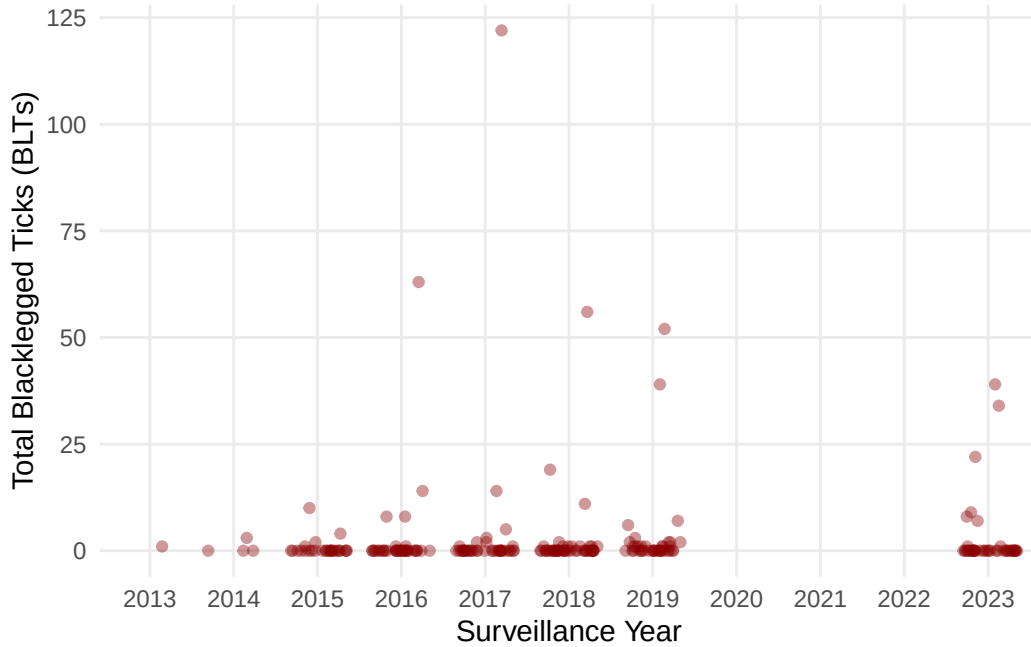


Figure 1: Adult and nymph ticks collected in each survey, 2013–2023; each dot is one site in one year.

Table 2 shows that larvae were rare, with just 22 being found in total. This tally comes from 4 surveys at 3 different sites (Anewen Greenbelt in 2017 and 2018, Highland Creek Park in 2017, and Cedar Ridge Park in 2018). Larvae hatch from eggs laid by fed adult females (Eisen et al. 2016), which could suggest that ticks were reproducing at those sites. Larvae are not tested and are not part of the total tick count, so we exclude them from the figures that follow.

2.7 Ticks are concentrated in the east

The majority of ticks were on the east side of Toronto (Figure 2). Sites that were east of longitude 79.25°W produced around 88% of all collected ticks. The Rouge Valley sites (five in total) produced 65% of the total ticks found (392 ticks). There are smaller clusters around the Toronto Islands, but most sites never recorded any ticks. Of all 84 sites surveyed, 52 had no ticks in any year.

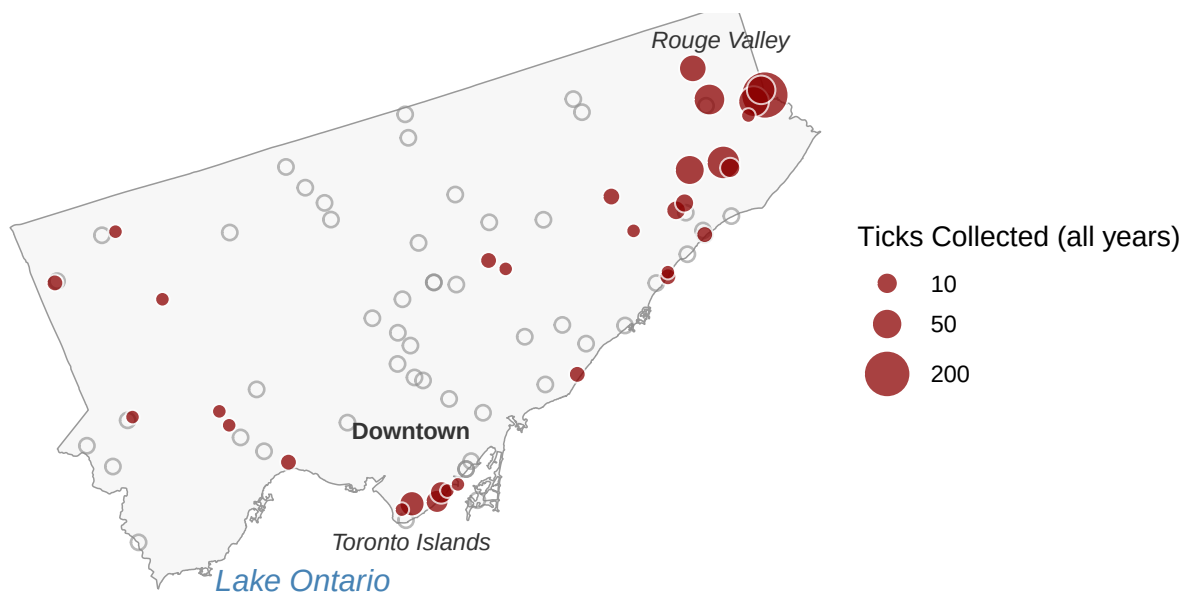


Figure 2: Surveillance sites across Toronto, sized by total ticks collected; hollow grey circles mark sites where no ticks were found.

Figure 3 showcases the top ten sites with the greatest number of ticks. Rouge Park recorded 209, and the Rouge Valley sites occupy five spots in the top 10, with Colonel Danforth Trail and Morningside Park also representing Toronto's east end. This illustrates the pattern that climate assigns a broad area where ticks can inhabit, but actual tick numbers within that region depend on woodland habitat and host densities (Eisen et al. 2016). Rouge Valley is a large woodland area, the type of habitat that would be expected to hold more ticks.

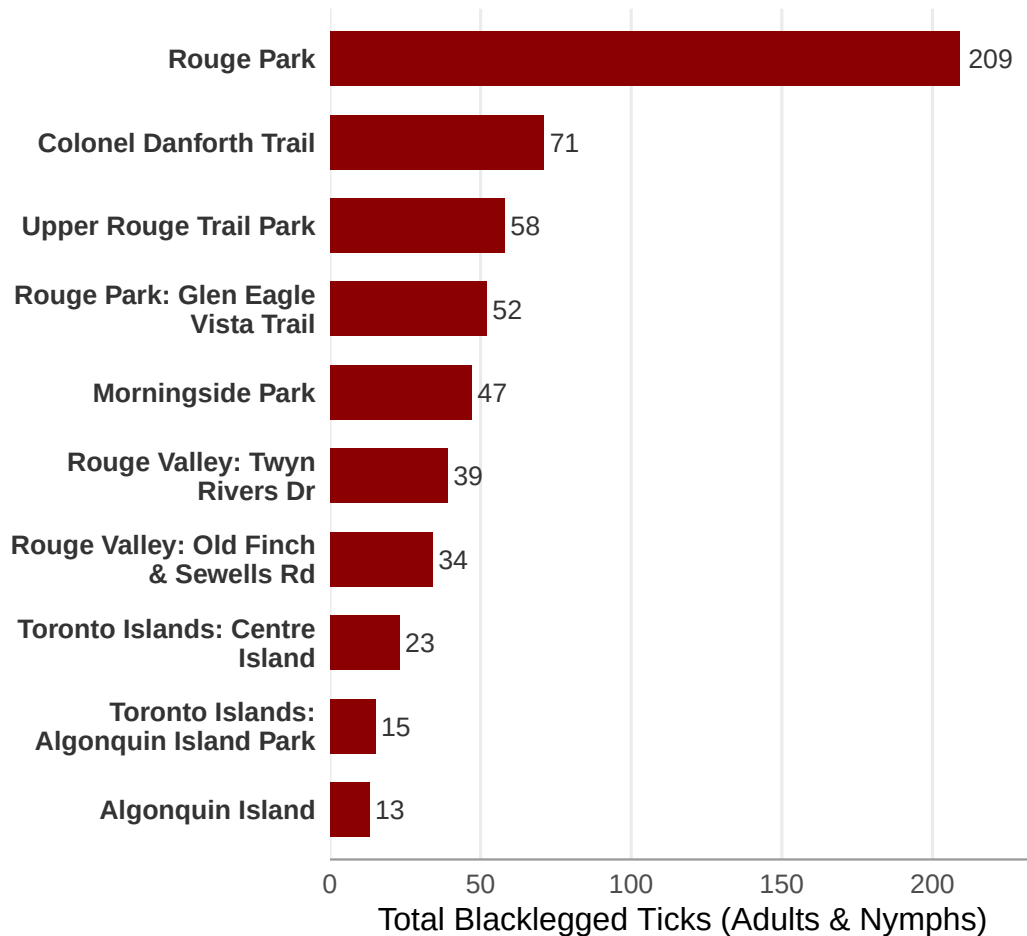


Figure 3: The ten sites with the most blacklegged ticks, summed across all surveillance years.

2.8 Infected ticks are concentrated in large catches

Of all 606 adults and nymphs collected, 35% or 211 ticks tested positive for the *Borrelia burgdorferi* bacterium, compared with 27.4% collected by dragging across Ontario in 2022 (Wafy et al. 2026). Most infected ticks were present in the small number of large catches

(Figure 4). The 13 surveys that found 10 or more ticks accounted for 82% of all ticks found but 96% of positive ones.

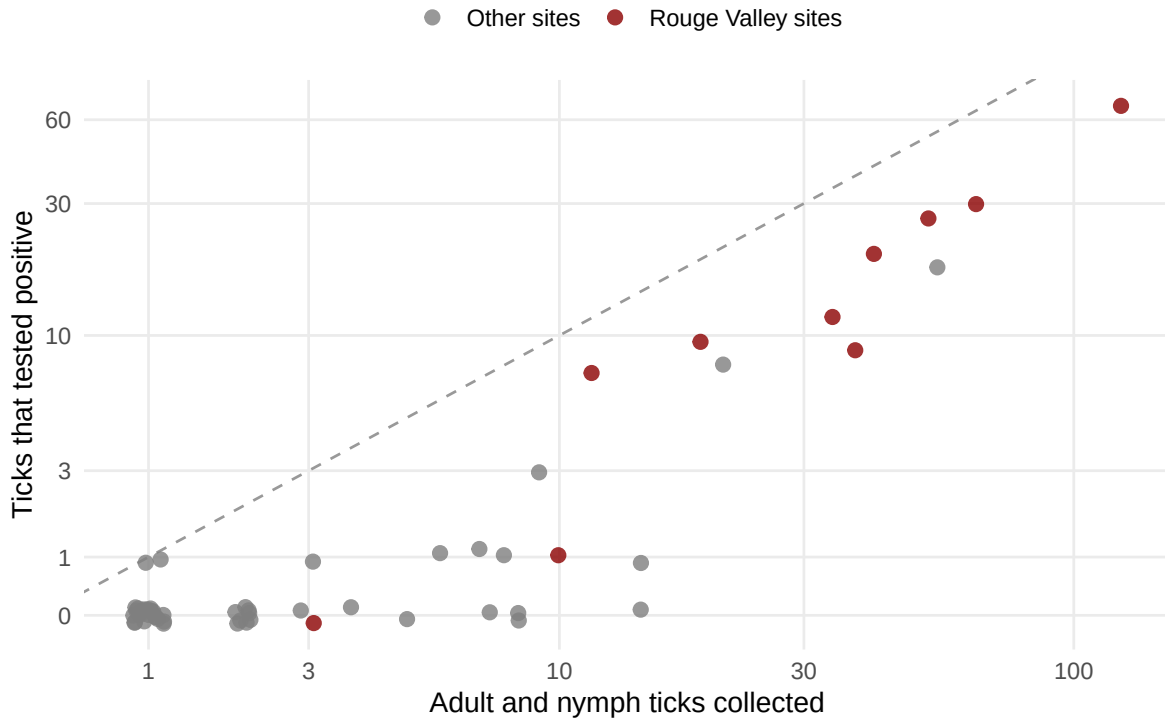


Figure 4: Ticks that tested positive against ticks collected, for the 57 surveys that found at least one tick.

The positivity percentage increased sharply from just under 6% in 2015 to between 29% and 43% in the years that followed (Table 2), but we cannot attribute this to climate change. This jump is largely from Rouge Park, where 29 of 63 ticks tested positive in 2016 alone. Outside the Rouge Valley, the data show no steady trend in this regard. This reflects the literature: the spread of blacklegged ticks in Canada has coincided with warming, but no change in exposure has yet been conclusively linked to climate change (Eisen et al. 2016).

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A Data cleaning

The raw data were downloaded from the Open Data Portal with `opendatatoronto` (Gelfand 2025), which was then saved before cleaning. Cleaning made four changes. Each site’s location is stored in the raw data as a text field in GeoJSON format. The first change was to extract the longitude and latitude from this text into two numeric columns. Second, the columns were renamed to consistent, readable names, and the original location field was dropped. Third, the count columns and year were converted to whole numbers. Fourth, only the columns used in this paper were kept. There were no rows that were removed.

Two issues were noted but not changed, so that the data match what Toronto Public Health published. Two sites, Upper Rouge Trail Park and Cedar Ridge Park, have slightly different coordinates in different years. The cleaned data keep both, and only the map in Figure 2 places each site at its first recorded location so that it appears once. There are also several sites that appear under more than one name, such as “Wards Island” and “Toronto Islands: Ward’s”, or “Guild Park” and “Guild Park and Gardens”. These were left as recorded, which is why we refer to 84 site names rather than 84 distinct places.

The cleaned data were checked with tests written using `testthat` (Wickham 2011), which confirm the number of rows and columns, that each year is a surveillance year, that site names are never missing, that the total count equals the adult and nymph count, that positive ticks never exceed ticks collected, and that every location falls within Toronto.